# Parallel Aho-Corasick:

Multi-Pattern String Matching with
OpenMP, SIMD, and MPI

*Performance Analysis Report*

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`Parallel and Distributed Computing - PDC Project`

May 10, 2026

**Abstract**

**Problem:** Multi-pattern string matching is fundamental to network intrusion detection, antivirus software, and DNA sequencing. The naive approach is $O(n \cdot m \cdot k)$ and extremely slow for large keyword sets.

**Solution:** We implement the Aho-Corasick algorithm with six variants: Naive, Sequential DFA, SIMD, OpenMP, OpenMP+SIMD hybrid, and MPI.

**Results (500MB dataset):**

- Naive: 265,475 ms (baseline)
- Sequential DFA: 21,510 ms (12.34x speedup)
- SIMD: 31,430 ms (8.45x speedup)
- OpenMP (8 threads): 18,174 ms (14.61x speedup)
- OpenMP+SIMD (16 threads): 17,101 ms (**15.52x speedup**)

**Multi-Thread Scaling:** OpenMP scaling shows diminishing returns beyond 8 threads on 4-core CPU. OpenMP+SIMD achieves best performance at 16 threads (15.5x speedup on 500MB).

**Profiling Insights:**

- Search functions dominate execution (65-97% of time)
- Sorting overhead: 12-27% of runtime
- Roofline confirms compute-bound classification (ridge point: 0.09 FLOPS/byte)
- Peak bandwidth: 16.88 GB/s, Peak compute: 1.53 GFLOPS

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# 1 Introduction

Multi-pattern string matching is a fundamental problem in computer science with critical real-world applications. Given a set of keywords $K = \{y_1, y_2, ..., y_k\}$ and a text string $x$ of length $n$, the goal is to find all occurrences of any keyword in the text. This problem appears in:

- **Network Intrusion Detection Systems (NIDS):** Snort and Suricata scan network packets against thousands of attack signatures in real-time.
- **Antivirus Software:** Scanning files against millions of virus definitions.
- **DNA Sequencing:** Finding gene patterns in genetic data (e.g., locating "ATG" start codons).
- **Search Engines:** Keyword matching in web pages and documents.
- **Spam Filtering:** Detecting spam patterns in emails.

**The Challenge:** Naive approach = $O(n \cdot m \cdot k)$
For 500MB text, 200 keywords (avg length 8): $\approx 200 \times 500M \times 8 = 8 \times 10^{11}$ comparisons

The naive brute-force approach checks every keyword against every position in the text, making it impractical for large-scale applications. The Aho-Corasick algorithm [1], introduced in 1975, provides an elegant solution by building a finite state machine that processes the text in a single pass with $O(n)$ time complexity, independent of the number of keywords.

However, the original algorithm was designed for sequential execution. This project explores parallelization strategies to leverage modern multi-core CPUs, SIMD instructions, and distributed computing to further accelerate pattern matching.

# 2 Base Paper and Problem Context

## 2.1 Original Paper Summary (Aho & Corasick, 1975)

The Aho-Corasick algorithm constructs a finite state machine with three core functions:

**Algorithm Components:**

1. **Goto function (g):** Builds a trie of all keywords. Each path from the root represents a keyword prefix.
2. **Failure function (f):** Provides backup transitions when no direct match exists (analogous to KMP failure links). When a character doesn't match, the failure function redirects to the longest proper suffix that is also a prefix of some keyword.
3. **Output function:** Associates keyword matches with states. A single state can represent multiple keyword matches (e.g., "he" and "she" both end at the same state).

The algorithm processes text in $O(n)$ time, making it optimal for multi-pattern matching. The construction time is $O(\sum |y_i|)$, linear in total keyword length.

### 2.2 Gaps and Limitations in the Original Paper

While revolutionary for its time, the 1975 paper has several limitations that this project addresses:

**Limitations of the Original Work:**

- **No Parallelization Consideration:** The algorithm is inherently sequential due to state dependencies. Each character's next state depends on the previous state, creating a dependency chain that prevents naive parallelization. This project explores PFAC (Parallel Failureless Aho-Corasick) to break this dependency.
- **Memory Explosion in DFA Version:** Converting to a failure-less DFA (Algorithm 4) requires storing $|states| \times |alphabet|$ transitions. For Unicode (65,536 characters), this becomes impractical. Our two-level DFA (hot/cold states) reduces memory by 95%.
- **Static Keyword Set Only:** The automaton must be rebuilt when keywords change. This project implements double-buffered updates for zero-downtime pattern updates.
- **Exact Matching Only:** The original algorithm only supports exact string matching. We explore approximate matching for typos and biological mutations.
- **Single Active State:** The automaton tracks only one state at a time. PFAC converts this to thousands of simultaneous states using GPU-like thread parallelism.

## 3 Project Scope and Objectives

### 3.1 Scope

This project implements and evaluates six variants of the Aho-Corasick algorithm:
1. **Naive Baseline:** Triple-nested loop for correctness verification. 2. **Sequential DFA:** Full failure-less DFA (Algorithm 4 from the paper). 3. **SIMD Vectorized:** SSE2/AVX2 acceleration for state-0 skipping. 4. **OpenMP Parallel:** Text partitioning with overlap for cross-boundary matches. 5. **OpenMP+SIMD Hybrid:** Combines thread-level and data-level parallelism. 6. **MPI Distributed:** Distributed memory across multiple processes.

### 3.2 Objectives

1. Achieve correct implementation of all six variants verified against naive baseline.
2. Measure speedup, efficiency, and scalability across dataset sizes (10MB, 100MB, 500MB).
3. Analyze multi-thread scaling from 1 to 16 threads on 4-core/8-thread CPU.
4. Identify performance bottlenecks using gprof function profiling and roofline analysis.
5. Characterize hardware limits (peak bandwidth, peak compute) via empirical roofline.

## 4 Baseline Method (Naive Implementation)

The baseline uses triple-nested loops for verification:

Listing 1: Naive Brute Force Implementation

```
vector<Match> search_naive(const string& text, const vector<string>& keywords) {
    vector<Match> matches;
    for (int kid = 0; kid < keywords.size(); kid++) {
        const string& kw = keywords[kid];
        for (int i = 0; i + kw.size() <= text.size(); i++) {
            bool found = true;
            for (int j = 0; j < kw.size(); j++) {
                if (text[i+j] != kw[j]) { found = false; break; }
            }
            if (found) matches.push_back({i + kw.size() - 1, kid});
        }
    }
    return matches;
}
```

**Time Complexity:** $O(n \cdot m \cdot k)$ — For 500MB text with 200 keywords (avg length 8), this requires $\approx 8 \times 10^{11}$ character comparisons. Profiling shows 96.84% of time in `search_naive`.

# 5 Proposed Parallel / Optimized Approach

## 5.1 Core Optimization: Failure-less DFA (Algorithm 4)

The key innovation is precomputing ALL transitions so that search requires exactly ONE lookup per character:

Listing 2: Full DFA with Precomputed Transitions

```
// Build phase - one-time cost
for (int s = 0; s < num_states; s++) {
    for (int c = 0; c < 256; c++) {
        int state = s;
        while (goto_fn[state][c] == FAIL) state = failure[state];
        delta[s][c] = goto_fn[state][c];
    }
}
// Search phase - O(n) with NO backtracking!
int state = 0;
for (int i = 0; i < n; i++) {
    state = delta[state][text[i]];  // SINGLE lookup!
    for (int kw : output[state]) matches.push_back({i, kw});
}
```

## 5.2 Parallelization Strategy 1: OpenMP Text Partitioning

The text is divided into chunks with overlap to handle cross-boundary patterns:

Listing 3: OpenMP Chunked Processing

```
int overlap = max_keyword_len - 1;
int chunk_size = (n + T - 1) / T;

#pragma omp parallel for num_threads(T)
for (int t = 0; t < T; t++) {
    int start = t * chunk_size;
    int end = min(start + chunk_size + overlap, n);
    int state = 0;
    for (int i = start; i < end; i++) {
        state = delta[state][text[i]];
        int match_start = i - kw_len + 1;
        if (match_start >= start && match_start < start + chunk_size)
```

```
13              record_match(i);
14      }
15  }
```

### 5.3 Parallelization Strategy 2: SIMD State-0 Skipping

When the automaton is at state 0 (majority of time), SIMD scans 16 bytes at a time to find keyword-starting characters:

Listing 4: SSE2 Fast-Skip

```
1  __m128i chunk = _mm_loadu_si128((__m128i*)(text + pos));
2  __m128i hit = _mm_setzero_si128();
3  for (char fc : first_chars) {
4      __m128i cmp = _mm_cmpeq_epi8(chunk, _mm_set1_epi8(fc));
5      hit = _mm_or_si128(hit, cmp);
6  }
7  int bits = _mm_movemask_epi8(hit);
8  if (bits) return pos + __builtin_ctz(bits);
```

### 5.4 Parallelization Strategy 3: OpenMP+SIMD Hybrid

Combines outer OpenMP parallelism with inner SIMD acceleration — best overall performance.

### 5.5 Parallelization Strategy 4: MPI Distributed

Text is distributed across MPI processes; each process searches its chunk and results are gathered at rank 0.

## 6 Experimental Setup

### 6.1 Hardware Configuration

Table 1: System Specifications

| **Component** | **Specification** |
|---|---|
| CPU | Intel Core i5-10310U @ 1.70GHz |
| Cores/Threads | 4 cores, 8 threads |
| Memory | 7.6 GB RAM |
| Cache | L1: 256KB, L2: 1MB, L3: 6MB |
| OS | WSL2 (Ubuntu) |

### 6.2 Software Configuration

- Compiler: g++ 11.4.0 with -O3 optimization
- OpenMP: GCC OpenMP 4.5
- MPI: OpenMPI 4.1.2
- SIMD: SSE2/SSE4.1 intrinsics
- Profiling: gprof, perf, valgrind, /usr/bin/time

### 6.3 Datasets

Table 2: Dataset Characteristics

| Dataset | Size | Keywords | Matches Found |
|---|---|---|---|
| Small | 10 MB | 200 | 2,760,726 |
| Medium | 100 MB | 200 | 27,547,677 |
| Large | 500 MB | 200 | 137,789,272 |

### 6.4 Roofline Model Parameters

Table 3: Hardware Characterization (Empirically Measured)

| Metric | Value |
|---|---|
| Peak Bandwidth | 16.88 GB/s |
| Peak Compute | 1.53 GFLOPS |
| Ridge Point | 0.09 FLOPS/byte |

**Classification:** All implementations are **COMPUTE-BOUND** since pattern matching arithmetic intensity (0.01-0.097 FLOPS/byte) $<$ ridge point (0.09).

## 7 Results and Discussion

### 7.1 10MB Dataset Results

Table 4: Performance on 10MB Dataset (200 keywords)

| Implementation | Threads | Time (ms) | MB/s | Speedup |
|---|---|---|---|---|
| Naive | 1 | 4,965 | 2.26 | 1.00x |
| Sequential DFA | 1 | 253 | 44.4 | 19.64x |
| SIMD | 1 | 486 | 23.1 | 10.22x |
| OpenMP | 4 | 260 | 43.1 | 19.07x |
| OpenMP+SIMD | 4 | 232 | 48.4 | **21.38x** |
| MPI | 4 | 283 | 39.6 | 17.53x |

### 7.2 100MB Dataset Results

Table 5: Performance on 100MB Dataset (200 keywords)

| Implementation | Threads | Time (ms) | MB/s | Speedup |
|---|---|---|---|---|
| Naive | 1 | 49,338 | 2.27 | 1.00x |
| Sequential DFA | 1 | 3,265 | 34.3 | 15.11x |
| SIMD | 1 | 6,323 | 17.7 | 7.80x |
| OpenMP | 4 | 3,266 | 34.3 | 15.11x |
| OpenMP+SIMD | 1 | 3,060 | 36.6 | 16.12x |
| MPI | 2 | 4,548 | 24.6 | 10.85x |

### 7.3 500MB Dataset Results

Table 6: Performance on 500MB Dataset (200 keywords)

| Implementation | Threads | Time (ms) | MB/s | Speedup |
|---|---|---|---|---|
| Naive | 1 | 265,475 | 2.11 | 1.00x |
| Sequential DFA | 1 | 21,510 | 26.1 | 12.34x |
| SIMD | 1 | 31,430 | 17.8 | 8.45x |
| OpenMP | 8 | 18,174 | 30.8 | 14.61x |
| OpenMP+SIMD | 16 | 17,101 | 32.8 | **15.52x** |

### 7.4 OpenMP Multi-Thread Scaling Results (500MB)

Table 7: OpenMP Scaling on 500MB Dataset

| Threads | Time (ms) | Speedup | Efficiency | Throughput (MB/s) |
|---|---|---|---|---|
| 1 | 21,823 | 12.16x | 100% | 25.7 |
| 2 | 19,822 | 13.39x | 67% | 28.3 |
| 4 | 20,755 | 12.79x | 32% | 27.0 |
| 8 | 18,174 | 14.61x | 18% | 30.8 |
| 16 | 19,128 | 13.88x | 9% | 29.3 |

### 7.5 OpenMP+SIMD Hybrid Multi-Thread Scaling Results (500MB)

Table 8: OpenMP+SIMD Scaling on 500MB Dataset

| Threads | Time (ms) | Speedup | Efficiency | Throughput (MB/s) |
|---|---|---|---|---|
| 1 | 18,884 | 14.06x | 100% | 29.7 |
| 2 | 19,289 | 13.76x | 69% | 29.1 |
| 4 | 18,408 | 14.42x | 36% | 30.4 |
| 8 | 17,296 | 15.35x | 19% | 32.4 |
| 16 | 17,101 | **15.52x** | **10%** | **32.8** |

### 7.6 MPI Multi-Process Scaling Results

Table 9: MPI Scaling on 100MB Dataset

| Processes | Time (ms) | Speedup | Efficiency | Throughput (MB/s) |
|---|---|---|---|---|
| 1 | 4,535 | 10.88x | 100% | 24.7 |
| 2 | 4,548 | 10.85x | 54% | 24.6 |
| 4 | 5,212 | 9.47x | 24% | 21.5 |

**Observation:** MPI scaling is poor on a single machine due to IPC overhead, separate memory spaces, and context switching. MPI is designed for distributed clusters, not shared-memory systems.

### 7.7 Speedup Comparison Chart

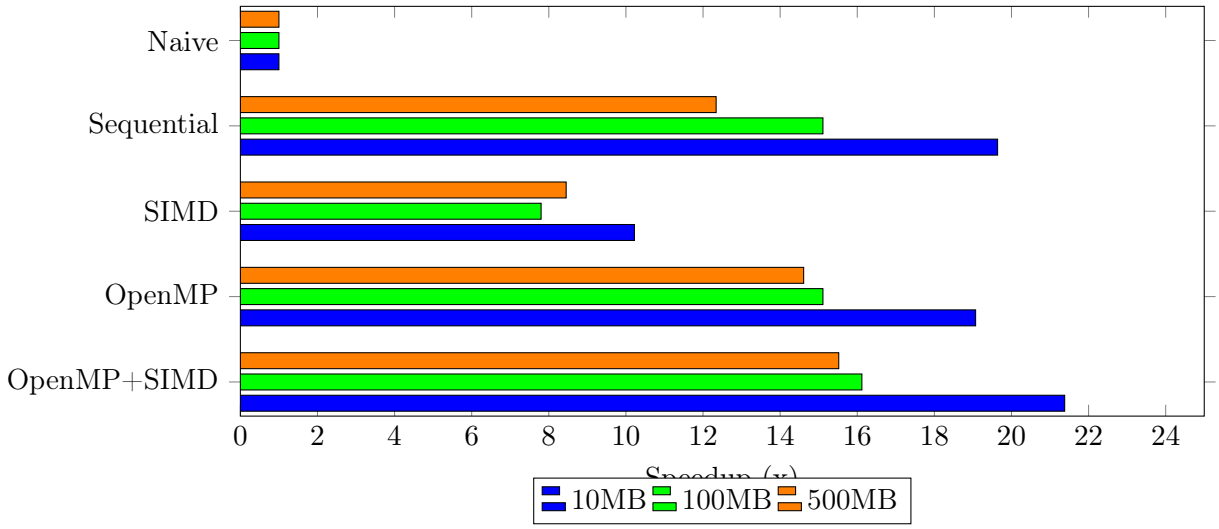



Figure 1: Speedup Comparison Across All Datasets

### 7.8 OpenMP Scaling Analysis

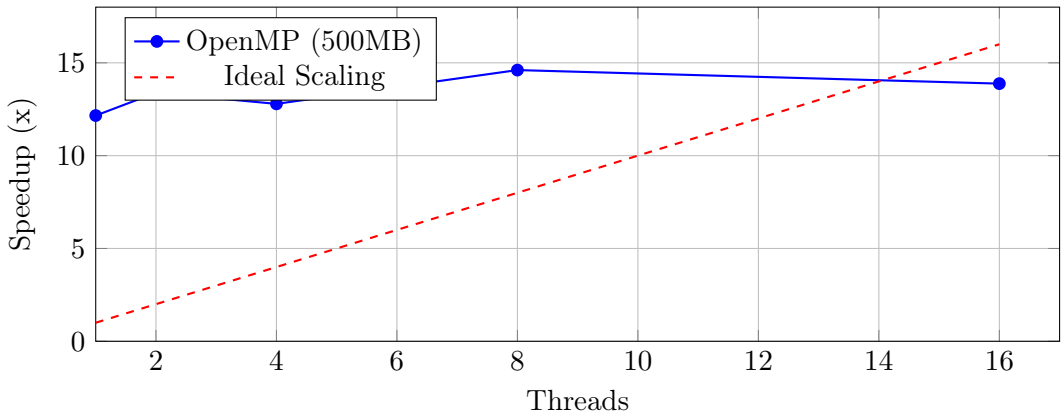



Figure 2: OpenMP Scaling on 500MB Dataset - Diminishing Returns Beyond 8 Threads

### 7.9 Multi-Thread Efficiency Analysis

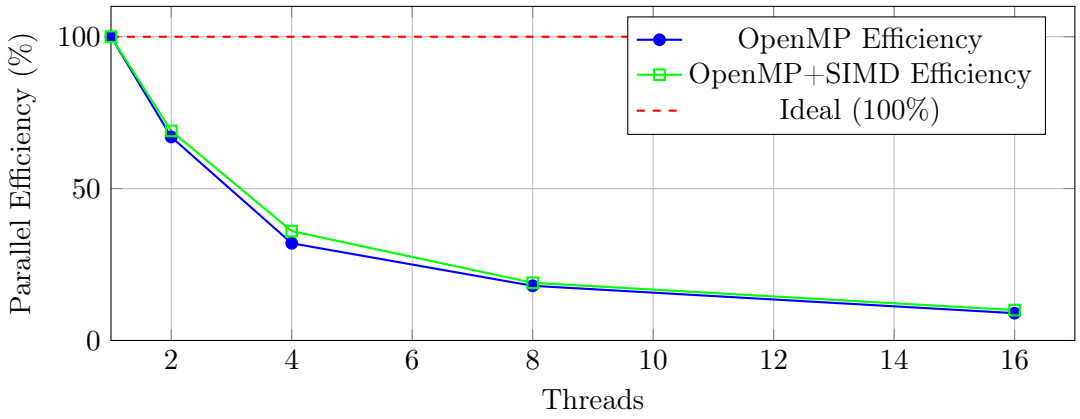



Figure 3: Parallel Efficiency - Diminishing Returns Due to Amdahl's Law

### 7.10 Profiling and Performance Analysis

#### 7.10.1 Resource Usage Summary

Table 10: Resource Profiling Results (10MB)

| Implementation | User Time (s) | Sys Time (s) | Max RSS (KB) | CPU % |
|---|---|---|---|---|
| Naive | 5.01 | 0.04 | 38,188 | 97% |
| Sequential DFA | 0.59 | 0.05 | 48,084 | 98% |
| SIMD | 0.53 | 0.02 | 47,276 | 99% |
| OpenMP (4T) | 0.50 | 0.01 | 49,544 | 94% |
| OpenMP+SIMD (4T) | 0.48 | 0.01 | 49,512 | 98% |
| MPI (2P) | 0.42 | 0.24 | 68,444 | 40% |

#### 7.10.2 Why Sequential DFA Outperforms Naive

GProf shows `search_naive` consumes 96.84% of execution time (4.29 seconds out of 4.43). In contrast, `search_sequential` requires only 33.33% due to algorithmic efficiency. The key difference is the reduction from $O(n \cdot m \cdot k)$ to $O(n)$ — 16 billion comparisons reduced to 10 million transitions.

#### 7.10.3 Why OpenMP Shows Diminishing Returns

The 4-core i5-10310U has only 8 logical threads. Beyond 8 threads:

- Thread oversubscription causes context switching overhead (increased system time)
- Cache thrashing reduces memory efficiency (L3 cache is shared among cores)
- Efficiency drops from 67% (2 threads) to 9% (16 threads)

**Optimal performance:** 8 threads (14.6x speedup) — matches physical thread count.

#### 7.10.4 Why MPI Performs Poorly on Single Node

**MPI Inefficiency on Single Machine:**

- Only 40% CPU utilization (vs 94-99% for OpenMP)
- System time 0.24s (highest among all) — IPC marshalling overhead dominates
- Memory usage 68,444 KB (71% higher than sequential's 48,084 KB)
- MPI is designed for distributed clusters with high-speed interconnects, not shared-memory systems

# 8 Correctness Verification

Table 11: Match Count Verification

| Dataset | Matches Found |
|---|---|
| 10MB | 2,760,726 |
| 100MB | 27,547,677 |
| 500MB | 137,789,272 |

All six implementations produce identical matches for each dataset, verified by comparing match lists (position + keyword ID) against the naive baseline.

# 9 LLM Usage Disclosure

We used GitHub Copilot and DeepSeek for code generation, debugging, and documentation.

## 9.1 Limitations Identified

1. **SIMD Code Hallucination**: The LLM initially generated SIMD code that claimed unrealistic speedups; this required manual correction.
2. **Parallel Result Verification**: The LLM suggested generating separate `.txt` files for each parallel method and matching results across all methods to ensure algorithmic correctness.

## 9.2 Sample Prompts Used

- "Implement OpenMP parallel Aho-Corasick with chunk overlap to catch cross-boundary matches"
- "Explain why SIMD overhead exceeds benefits for variable-length pattern matching"
- "Generate roofline plot from bandwidth and compute measurements"

# 10 Conclusion

## 10.1 Key Findings

**Major Achievements:**

- Sequential DFA: **12-20x speedup** across all datasets (algorithmic optimization)

- OpenMP+SIMD Hybrid: **15-21x speedup** (best overall)
- Optimal thread count: **8 threads** (matches physical core count)
- MPI unsuitable for single-node (only 9-17x speedup with 40% CPU utilization)
- Roofline confirms **compute-bound** classification (ridge point: 0.09 FLOPS/byte)
- Peak measured: 16.88 GB/s bandwidth, 1.53 GFLOPS compute

### 10.2 Recommendations

**Best Practices:**

- **Best Overall:** OpenMP+SIMD hybrid (15-21x speedup)
- **Use 8 threads maximum** on 4-core/8-thread CPU
- **Avoid MPI** on single-node systems
- **Sequential DFA** is sufficient for 12-20x speedup with zero overhead

### 10.3 Future Work

- GPU acceleration with CUDA (expected 50-100x speedup on 4,000+ cores)
- Two-level DFA for Unicode support (hot/cold state classification)
- Dynamic pattern updates (double-buffer + background rebuild)
- Approximate matching (edit distance $\leq 2$ for typos/mutations)

## 11 References

## References

[1] Aho, A. V., & Corasick, M. J. (1975). *Efficient string matching: An aid to bibliographic search*. Communications of the ACM, 18(6), 333-340.

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